

VITHUSHA PATHMANATHAN

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SUMMARY

Final-year Software Engineering undergraduate at SLIIT with hands-on industry experience building production-grade full-stack and AI-powered applications at Nilavitech. Proficient in Java (Spring Boot), Python (FastAPI), and React (TypeScript), with additional experience in React Native, Next.js, AI/ML integrations (LangChain, Groq, Google Gemini), cloud platform (AWS), and conversational AI systems. Seeking an Associate Software Engineer role to apply strong full-stack development and problem-solving skills in building scalable, real-world solutions.

TECHNICAL SKILLS

Languages	Java, JavaScript, TypeScript, Python, C, C++, PHP, Kotlin, HTML, CSS
Frameworks	Spring Boot, FastAPI, React.js, Next.js, React Native, Node.js, Express.js
AI / Cloud / DevOps	LangChain, Groq AI, Google Gemini, Dialogflow CX, OpenAI API, AWS, Docker
Databases	PostgreSQL, MongoDB, Redis, Appwrite
Tools	GitHub, Postman, Figma, VS Code, IntelliJ IDEA, Android Studio, Swagger, Jupyter Notebook, CI/CD (GitHub Actions)
Concepts	REST API Design, RBAC, Microservices, OOP, JWT Auth, Image Processing
AI Libraries	OpenCV

WORK EXPERIENCE

Nilavitech | Intern Software Engineer

Sep 2025 – Present

Tiger Express | Cross-Border Logistics Platform

Spring Boot, React, TypeScript, PostgreSQL, AWS

- Built a cross-border logistics platform (Australia ↔ Sri Lanka & India) with sea/air freight options, a dynamic pricing engine calculating costs by weight and item type, and role-based access control (RBAC) for Admin, Agent, and Client roles
- Designed and deployed RESTful APIs via AWS API Gateway with Swagger UI documentation; built the frontend in React with TypeScript [Tiger Express](#)

MedTrace AI Web | Full-Stack Clinical Intelligence Platform

Next.js, TypeScript, FastAPI, Python, PostgreSQL, Google Gemini, JWT, Docker

- Built an AI-powered medical document pipeline: PDF/text ingestion (pdfplumber) → entity extraction and clinical summaries via Google Gemini with rule-based fallback → structured storage in PostgreSQL
- Implemented patient health timelines, rule-based risk detection, and alert-review APIs, fully integrated with the frontend

AI Receptionist | Multi-Channel Conversational AI System

FastAPI, Dialogflow CX, Google STT/TTS, Twilio, React, PostgreSQL, Redis

- Built a multi-channel AI receptionist handling voice calls, web chat, and SMS using Dialogflow CX, Google Speech-to-Text/Text-to-Speech, and Twilio with WebSocket support for real-time conversations
- Integrated Google Calendar API for appointment sync and OpenAI/Gemini for NLP-driven booking, rescheduling, and cancellation handling

Hospital Patient Portal & Secure Access Platform

FastAPI, SQLAlchemy (async), PostgreSQL, Pydantic, JWT, Twilio, Next.js, TypeScript, React, Tailwind CSS, Docker, Gemini AI

- Developed a secure multi-role healthcare platform for patient registration, specialist discovery, and appointment management with role-based access for patients, doctors, and admins
- Built a consent-driven medical record access workflow with patient approval, OTP verification, time-limited access sessions, and audit-focused security controls for regulatory compliance

ACADEMIC PROJECTS

Microservice-Based Event Booking Platform

[\[GitHub link\]](#)

Node.js, Express.js, MongoDB, REST APIs, Microservices, React

- Developed the Event Service in a microservices-based booking platform, handling event management (create, update, availability) and inter-service communication with Booking and User services via REST APIs and MongoDB
- Contributed to a distributed microservices architecture (User, Event, Booking, Review services) for an event booking platform

Tax-Trail - AI-Powered Tax Transparency Mobile App

[\[GitHub link\]](#)

React Native, TypeScript, Expo, LangChain, Groq AI, Node.js, Appwrite, Zustand

- Built a mobile app aligned with UN SDG 10 (Reduced Inequalities), integrating LangChain and Groq for AI-powered anomaly detection in financial data, trend forecasting, and automated executive summary generation
- Developed a real-time multi-currency converter (150+ currencies) with live exchange rate charts; used Appwrite for Google OAuth/database and Zustand for state management

GreenLoop - Smart Waste Management System

[\[GitHub link\]](#)

MERN Stack, Redux Toolkit, Stripe, Firebase, Tailwind CSS, Chart.js, QR Code

- Team project integrating Stripe payments, Firebase OAuth, Redux Toolkit, and a real-time Chart.js analytics dashboard across 4 independently developed feature modules, with conflict detection, Factory Design Pattern, admin approval workflow, and QR-based collection verification

EDUCATION

BSc (Hons) in Information Technology - Specializing in Software Engineering

Sep 2022 – Present

Sri Lanka Institute of Information and Technology (SLIIT) | Final Semester

G.C.E. Advanced Level | Jaffna Vembadi Girls' High School

2019

CERTIFICATIONS

- Python for Beginners - University of Moratuwa
- Java Programming - Great Learning Academy
- React JS - Great Learning Academy
- UI/UX for Beginners - Great Learning Academy
- NVQ Level 03 - Computer Application Assistant (CAA)
- Diploma in Information Technology with Distinction - DMI Computer Education
- Diploma in English - College of English and IT Skills Development

EXTRA ACTIVITIES

- Secretary - Media Club, SLIIT Northern Uni - 2023
- Library Day Speech Competition - School Level

REFERENCES

Ms. Bhagya Nathali Silva
Assistant Professor -at - SLIIT
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Ms.Uthpala Samarakoon
Senior Lecturer at - SLIIT
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